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The Impact of the "Steroid Crash" on Athletes' Health-Related Quality of Life: Psychosocial Consequences of Anabolic-Androgenic Steroid Withdrawal

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Abstract

Introduction. The use of anabolic-androgenic steroids (AAS) in strength sports has well-documented somatic effects; however, the withdrawal phase remains marginalized by sports medicine. Discontinuation of exogenous androgens triggers profound suppression of the hypothalamic-pituitary-gonadal (HPG) axis. This collapse drastically reduces athletes' quality of life, manifesting through severe depressive states and exacerbated body image disorders.

Objective. This review synthesizes current scientific knowledge on the psychosomatic consequences of the AAS withdrawal syndrome (the “steroid crash”). Emphasis is placed on the pathophysiology of hypogonadism, the psychological impact of muscle loss, systemic barriers to seeking professional help, and an analysis of self-administered pharmacological strategies (Post-Cycle Therapy – PCT).

Methodology. A structured search of databases, including PubMed and Scopus, was conducted covering literature up to mid-2026. Keywords included: “AAS withdrawal syndrome,” “post-cycle therapy,” “major depressive disorder,” “muscle dysmorphia,” and “HPG axis suppression.” Priority was given to systematic reviews and clinical trials detailing neuroendocrine mechanisms, suicide risk, and psychosocial factors among former users.

Conclusions. The AAS withdrawal phase constitutes a critical threat to athletes' well-being, associated with a high risk of clinical depression and active suicidal ideation. A lack of standardized medical guidelines, deep-rooted stigmatization, and distrust of medical staff force athletes into experimental self-treatment (informal PCT). Improving care requires shifting from moralizing to a harm-reduction model that combines endocrinological support with early psychiatric intervention.

Keywords: AAS withdrawal syndrome, anabolic-androgenic steroids, major depression, muscle dysmorphia, Post-Cycle Therapy, sports medicine

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1. Introduction

Modern strength sports and bodybuilding are inextricably linked to the pursuit of pushing the body's physiological limits, which often leads to the use of anabolic-androgenic steroids (AAS) [1, 2]. However, while the scientific literature and public debate extensively document the performance-enhancing and physical effects of doping, researchers rarely focus on the critical moment of discontinuing its use [3]. The withdrawal phase, colloquially referred to in the sports community as a “steroid crash,” is in fact a full-blown neuroendocrine withdrawal syndrome [4]. From the perspective of the topics covered in the journal *Quality in Sport*, this period appears to be one of the most drastic—and at the same time, one of the least studied—factors degrading athletes' overall quality of life (Health-Related Quality of Life—HRQoL). Sudden withdrawal of exogenous androgens, combined with a blocked hypothalamic-pituitary-gonadal (HPG) axis, not only undermines the physical condition that

has been built up but, above all, strikes at the very foundations of mental health [4, 5]. In this study, we critically evaluate the withdrawal phase, highlighting it not just as a transitional period, but as a direct and immediate threat to the athlete's welfare. The article demonstrates that the consequences of discontinuing AAS use extend far beyond the medical issue of transient hypogonadism, encompassing a broad spectrum of psychosomatic pathologies—ranging from severe muscle dysmorphia, through severe clinical depression, to the presence of active suicidal thoughts [4, 6]. Furthermore, this study attempts a critical assessment of systemic barriers—including deep-seated stigmatization and the lack of standardized therapeutic guidelines (Post-Cycle Therapy)—which force athletes to cope with this crisis in dramatic isolation [6, 8]. Understanding these multidimensional mechanisms is essential for medical and psychological care in sports to move away from a model based on systemic stigmatization and become a tool for harm-reduction-oriented intervention and the effective protection of athletes' health and lives.

2. Pathophysiology of withdrawal syndrome: suppression of the HPG axis and hypogonadism

Ending an extended cycle of anabolic-androgenic steroids plunges the athlete's body into a severe neuroendocrine crisis. This physiological response is based on pharmacological negative feedback. Prolonged administration of exogenous androgens at supraphysiological doses leads to the continuous saturation of both androgen and estrogen receptors (due to peripheral aromatization). This constant stimulation of the receptors effectively inhibits the release of gonadotropin-releasing hormone (GnRH) from the hypothalamus and of LH and FSH from the anterior pituitary. Without these essential biological signals, testosterone synthesis in the testes ceases entirely, leading to clinical hypogonadotropic hypogonadism [1, 2]. Although synthetic hormones are rapidly cleared from the systemic circulation after doping is discontinued, the hypothalamic-pituitary-gonadal axis does not immediately return to normal function; rather, it remains profoundly suppressed. Consequently, the athlete must cope with a sudden and drastic drop in circulating androgen levels, often plummeting to values significantly lower than those typically seen in elderly, clinically hypogonadal men.

The time required for this complex neuroendocrine system to recover is highly variable. It depends intrinsically on the individual's history of doping use—specifically regarding dosage, cycle duration, the biological half-lives of the injected esters, and the simultaneous use of multiple substances. Although some longitudinal studies suggest that serum testosterone levels may return to reference ranges within 3 to 10 months, clinical observations indicate that

fewer than half of patients achieve full, sustained hormonal normalization within this timeframe [7]. Such an unpredictably prolonged hypogonadal state can severely derail an athlete's training programming and overall competitive timeline. Consequently, the ensuing physical stagnation often exacerbates the individual's frustration, creating a compounding effect that impedes both somatic and psychological recovery. During this crucial period, men often experience severe testicular atrophy, erectile dysfunction, significant loss of libido, and temporary or potentially permanent infertility [1, 2]. This severe atrophy results from the prolonged absence of LH and FSH, which directly starves the Leydig cells (responsible for testosterone synthesis) and Sertoli cells (responsible for spermatogenesis). This deterioration of testicular tissue indicates that, even once pituitary signaling is restored, the shrunken gonads often do not respond, which can extend the duration of clinical hypogonadism.

However, it is not merely the physiological degradation but the severe psychological costs of a "steroid crash" that pose the greatest threat to an athlete's overall quality of life. In the scientific literature, this post-cycle phenomenon is increasingly described as a severe clinical triad consisting of hormonal hypogonadism, distressing somatic symptoms, and severe depressive and anxiety symptoms [1, 3, 4]. During the withdrawal phase, athletes often experience depressed mood, persistent fatigue, lethargy, severe anhedonia, and frequently, active suicidal thoughts along with intense cravings for steroids. From a neuroscientific perspective, these symptoms are closely connected to sudden disruptions in key neurotransmitter systems in the brain, especially serotonin and dopamine, which are essential for maintaining mood stability and regulating the reward system. The sudden absence of neuromodulatory androgens heavily destabilizes these pathways. However, it must be emphatically stressed that there remains a significant research gap: current knowledge regarding neurochemical changes following AAS discontinuation is based primarily on clinical inferences drawn from the symptoms of hypogonadism, as there is a lack of hard, empirical measurements of neurotransmitter levels or neuroimaging data in the human brain during this specific, high-risk phase [1, 3].

The sports community's desperate response to this dramatic decline in physical performance and psychological well-being is the autonomous implementation of so-called Post-Cycle Therapy (PCT). These informal protocols seek to pharmacologically trigger a faster restart of the dormant HPG axis using off-label drugs. This treatment usually includes human chorionic gonadotropin (hCG), which mimics LH to directly stimulate dormant Leydig cells. It also involves selective estrogen receptor modulators (SERMs) and aromatase inhibitors (AIs) to help reduce the inhibitory effects of circulating estrogens on the pituitary gland. Survey data

strongly suggest that over half of AAS users try these self-administered PCT protocols, often reporting a 50–60% decrease in the severity of physical withdrawal symptoms and suicidal thoughts. However, from the perspective of evidence-based medicine (EBM), there is a clear lack of reliable randomized controlled trials (RCTs) with control groups that would unequivocally confirm the clinical efficacy, optimal dosing, and long-term safety of these protocols [6, 9]. As a result, athletes, driven by desperation and a lack of systemic medical support, often conduct high-risk pharmacological experiments on themselves, attempting to alleviate life-threatening depressive states using complex chemical methods lacking a solid, peer-reviewed scientific basis [6, 8].

3. Psychopathology of the Withdrawal Phase: Clinical Depression and Suicide Risk

Stopping anabolic-androgenic steroids has effects that go well beyond just a temporary drop in physical performance; it can seriously impact athletes' mental health. From a clinical point of view, what athletes experience isn't just a short-term slump but a full withdrawal syndrome. The scale of this problem is extremely widespread—survey analyses indicate that as many as 95% of men attempting to stop doping experience negative symptoms [8]. The predominant symptom, reported by nearly three-quarters of athletes, is a drastic decline in mood [5, 8]. However, we are not talking here solely about temporary sadness or a decline in training motivation. Systematic literature reviews point to the co-occurrence of severe anhedonia, insomnia, body image disorders, and a strong craving for steroids that torments the athlete [3].

In a significant group of users, this condition worsens, progressing into serious psychiatric disorders. Studies based on structured clinical interviews show that, within just three months of stopping injections, a significant percentage of athletes develop major depressive disorder that meets rigorous diagnostic criteria [4]. Furthermore, statistics confirm that individuals with a history of androgen use exhibit a higher overall incidence of clinical depression compared to the general population that has never used performance-enhancing drugs [4, 5]. The most alarming—and at the same time the most frequently marginalized—aspect of the withdrawal phase, however, is the emergence of suicidal thoughts and tendencies. The medical literature explicitly classifies these as an integral, clinically significant component of withdrawal syndrome, driven by pharmacologically induced, profound hypogonadism [4, 5].

Desperate attempts to preserve their well-being are best illustrated by the fact that athletes resort to informal Post-Cycle Therapy protocols not only to restore their physiology

but also to survive mentally—reporting that self-administered hormonal interventions allow them to reduce the intensity of suicidal thoughts by half [8]. This dramatic psychological state, characterized by unimaginable suffering and a sense of loss of control, creates a dangerous vicious cycle. Deep depression, combined with psychological compulsion, becomes the main factor leading to a relapse and a return to the steroid cycle [4]. From the perspective of quality of life (QoL), the SAA withdrawal phase absolutely cannot, therefore, be treated as a routine process of restoring endocrine balance. It is a state of heightened psychiatric risk in which the athlete—often struggling with the problem in complete isolation—absolutely requires professional monitoring and support [4, 6].

4. The Psychological Costs of Muscle Loss: The Role of Muscle Dysmorphia

It is important to remember that the drama of the withdrawal phase unfolds not only at the biochemical level but also—and for many athletes, above all—in front of the mirror. The loss of hard-earned muscle mass and changes in body composition are among the strongest triggers of a psychological crisis, closely linked to body dysmorphic disorder (BDD), and in particular to a specific subtype known as muscle dysmorphia (bigorexia) [10]. In the literature, this disorder is described as a persistent, often obsessive belief that one is “too small” or “not muscular enough,” even when the athlete’s physique objectively demonstrates impressive musculature. During a steroid cycle, exogenous androgens not only promote muscle growth but also artificially boost self-esteem, masking deep-seated insecurities [10].

The reality check after discontinuing performance-enhancing drugs can be extremely brutal. The physiological decline in muscle volume and “fullness” abruptly exposes and radically intensifies dormant body dysmorphic symptoms [10]. For an athlete whose identity and self-esteem have been entirely built on a muscular physique, the widening gap between a deteriorating physique and an internalized ideal becomes a source of immense suffering. This psychological blow triggers an avalanche of destructive phenomena: athletes fall into the trap of compulsive training, drastic dietary restrictions, and social isolation, avoiding exposure of their bodies due to paralyzing shame and a sense of failure [10, 11, 12].

Clinical studies unequivocally indicate that clinically significant body dissatisfaction greatly increases the risk of serious depressive episodes [13]. In bodybuilders and strength athletes, muscle dysmorphia can negatively interact with hormonal mood disorders, making the situation even more challenging [4, 10]. This phenomenon leads to a vicious cycle in which weight loss following AAS discontinuation triggers self-loathing, which in turn exacerbates anxiety, neuroticism, and isolation. This picture clearly illustrates that an athlete’s quality of

life deteriorates significantly during the post-cycle phase. At the same time, it confirms the urgent need for a change in approach in sports medicine—routine screening for body dysmorphic disorder and monitoring of patients' psychological responses to muscle loss should be integral parts of care during the doping withdrawal process [10].

5. Social and systemic barriers to seeking specialized help

Despite a drastic decline in overall quality of life and an escalating psychiatric risk during the withdrawal phase, only about one-third of anabolic-androgenic steroid users ultimately decide to seek professional medical help [14, 15]. This alarming statistic stems from deeply rooted systemic and psychosocial barriers, among which the phenomenon of multidimensional stigmatization stands out as a primary deterrent [16]. In the highly competitive strength sports community, which is largely dominated by rigorous norms of traditionally understood masculinity and stoicism, admitting to a deterioration in mental health or emotional instability is often incorrectly equated with unacceptable personal weakness [17, 18]. Additionally, the use of performance-enhancing drugs itself carries a heavy social stigma of deviance and moral failure, further discouraging open dialogue.

This double burden of psychological shame is heavily compounded by a very real, tangible fear of severe disciplinary and legal consequences. The ongoing criminalization of the possession and distribution of AAS gives rise to a complex phenomenon described in the scientific literature as “politicogenic drug effects” [19, 20]. Elite and amateur athletes alike are often paralyzed by the fear that being completely honest in a doctor’s office will lead to an unintentional breach of patient confidentiality and a permanent entry in their official medical records. They fear that such disclosures could consequently ruin their athletic careers, result in their immediate removal from a team, trigger anti-doping sanctions, or even provoke direct intervention by law enforcement or social services [15, 21, 22].

A third, equally significant barrier is a widespread, systemic lack of trust in the specific competence of conventional medical personnel. Among AAS users, there is a prevailing belief that general primary care physicians and even psychiatrists lack sufficient practical knowledge of sports endocrinology, supposedly rendering them unable to offer empathetic or effective assistance during a post-cycle crash [6, 15]. As a result, instead of seeking safe, evidence-based clinical interventions, athletes instinctively resort to self-medication, heavily relying on so-called “bro-science”—informal, anecdotal advice gathered from coaches, fellow athletes, and unregulated, anonymous online forums [6, 15, 23]. These underground community support networks, while initially providing the patient with a comforting but false sense of belonging

and understanding, ultimately prove completely inadequate and dangerous in the face of full-blown clinical depression or active suicidal thoughts. Being trapped in this hermetic, echo-chamber information bubble turns the withdrawal phase into a period of extreme systemic isolation, which critically prevents early detection of psychiatric risks and significantly exacerbates the rapid decline in the athlete's overall quality of life.

6. Pharmacological Strategies and Post-Cycle Therapy Protocols vs. the Harm Reduction Paradigm

Despite the enormous scale of the problem and the dramatic decline in athletes' quality of life, modern medicine remains largely helpless in the face of withdrawal syndrome, lacking standardized, evidence-based treatment protocols. In the absence of guidelines, desperate athletes try to salvage what remains of their physiology and prevent a mental breakdown by resorting on their own to informal Post-Cycle Therapy protocols that adapt medications routinely used to treat infertility and clinical hypogonadism [6, 8].

Understanding this phenomenon requires an examination of its pharmacological foundations. The cornerstone of most protocols is human chorionic gonadotropin. Acting as a direct analog of luteinizing hormone, it bypasses the hypothalamus and pituitary gland—which are blocked after a cycle—to directly stimulate Leydig cells in the testes to resume testosterone production and prevent their atrophy [24]. To permanently restart the HPG axis, therapy additionally relies on selective estrogen receptor modulators, such as clomiphene. Their mechanism involves precisely blocking estrogen receptors in the hypothalamus, thereby eliminating negative feedback and forcing the body to release endogenous gonadotropin-releasing hormone [1, 25]. This therapy is often supplemented with aromatase inhibitors, which enhance the signal stimulating the pituitary gland [1, 2].

From the perspective of users' subjective experiences, this complex combination therapy appears highly effective. Extensive survey data indicate that self-administered pharmacotherapy can reduce the severity of physical withdrawal symptoms, steroid cravings, and—most importantly from a psychiatric perspective—suicidal thoughts by approximately 50 to 60 percent [8]. For many people, therefore, informal PCT is not merely an attempt to regain physical fitness, but a desperate struggle to survive a mental health crisis. Unfortunately, from an evidence-based medicine perspective, these practices remain off-label experimentation [6, 8]. There is a lack of rigorous randomized controlled trials to verify the efficacy, optimal dosing, or safety profile of these agents in treating withdrawal syndrome following SAA [6,

24]. Current knowledge is based mainly on extrapolating results from male infertility therapy and on the aforementioned self-reports from athletes.

As a result, there is a profound lack of consensus within the medical community itself. Some specialists opt for immediate initiation of hormone therapy. In contrast, others recommend a “wait-and-see” strategy, which involves passively waiting for the body to recover spontaneously. This process can take anywhere from six to over a dozen months [2, 15]. From a psychological perspective, leaving a patient with severe hypogonadism, anhedonia, and active suicidal tendencies without intervention for such a long period is an extremely dangerous and unethical situation. For this reason, the latest guidelines, based on the harm-reduction paradigm, emphasize a complete shift in approach. Instead of focusing exclusively on restoring reference testosterone levels, they advocate for the necessity of early, interdisciplinary support. The key lies in a holistic, non-judgmental approach to the patient, encompassing acute pharmacotherapy for severe depression and psychosocial interventions, including cognitive-behavioral therapy (CBT) targeted at treating muscle dysmorphia [4, 6]. Effectively alleviating the psychological costs of doping withdrawal requires recognizing that the clinician is not merely dealing with a patient with a temporary endocrine disorder, but with a person in a profound identity crisis who requires immediate intervention to preserve their quality of life.

7. Conclusions

An analysis of the relevant literature leads to the unequivocal conclusion that the withdrawal phase from anabolic-androgenic steroids represents one of the most marginalized crises in the field of modern sports medicine. A problem that has so far been reduced to transient disorders of the hypothalamic-pituitary-gonadal axis and a decline in physical performance is, in reality, a profound breakdown of psychosocial well-being. A drastic drop in androgen levels, combined with a rapid loss of hard-earned muscle mass, creates a dangerous foundation for the development of full-blown clinical depression, the exacerbation of body dysmorphic disorder, and the emergence of active suicidal thoughts [4, 5, 10]. The systemic isolation of athletes dramatically exacerbates this situation. Fear of stigmatization, concern about disciplinary consequences, and a widespread lack of trust in medical personnel who lack the necessary expertise mean that patients in the crash phase most often remain outside the system of professional care [16, 17]. The resort to self-medication and informal, scientifically unsupported Post-Cycle Therapy protocols highlights the complete failure of the current care model [6, 8]. To meaningfully improve the quality of life of this specific group, a fundamental paradigm shift is necessary—from moralizing and an exclusive focus on endocrine parameters

to an interdisciplinary, harm-reduction-based approach. Routine monitoring of mental health, including early detection of muscle dysmorphia and suicidal tendencies, should become an absolute standard in the care of athletes completing a pharmacological cycle. Only the integration of endocrinological care with specialized psychiatric and psychotherapeutic support can prevent the tragic consequences of withdrawal syndrome and effectively protect athletes' lives [4, 6, 15].

Disclosure

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